

Competition Mathematics

Օլիմպիադային մաթեմատիկա

COURSE CODE	MTH-COMP
PROGRAM	Competition Mathematics · Օլիմպիադային մաթեմատիկա
LEVEL	Grades 5–12 · three levels by placement, not by age
DURATION	72–108 instructional hours per level (approx. 60–90 lessons)
PREREQUISITE	Placement by assessment. Level I requires Grade 5 arithmetic fluency; Level II requires Algebra I; Level III requires Algebra II and a qualifying competition score.
LEADS TO	The next competition level, or continued olympiad training

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Competition Mathematics is a different subject from school mathematics, and Վարժարան teaches it as such. Where the school curriculum asks a student to apply a known method to a familiar problem, competition mathematics asks them to find a method for a problem they have never seen. The course develops the four classical olympiad areas — algebra, combinatorics, geometry, and number theory — together with the habits of mind that distinguish a competitor: systematic exploration, pattern recognition, willingness to sit with a hard problem, and the discipline of writing a complete argument. It runs at three levels aligned to the AMC 8, AMC 10/12, and AIME progression.

TEACHING APPROACH

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Competition training is often reduced to drilling past problems by topic, which produces students who recognize familiar problems and freeze on unfamiliar ones. Վարժարան trains problem-solving explicitly instead: students are taught named strategies — specialize, generalize, work backwards, look for invariants, consider extremes, count two ways, introduce symmetry — and are required to name the strategy they are attempting before they attempt it. Sitting with an unsolved problem is treated as the core activity rather than as failure, and lessons deliberately include problems the student will not solve within the hour. The teacher's role is to ask the question that unlocks the student's own next step, not to demonstrate a solution. Students also keep a solution journal, because writing a clean argument is a separate skill from finding it and is what olympiad grading actually rewards.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Persists on a problem for twenty minutes without asking for help	The single most predictive indicator. Content can be taught; tolerance for productive struggle must be built deliberately.
Has strong arithmetic and algebraic fluency for their level	Competition problems assume computation is free; a student computing slowly cannot explore.
Enjoys puzzles and unfamiliar problems rather than finding them threatening	Determines whether the course will be motivating or discouraging; discussed with the parent before enrollment.
Can write a clear explanation of how a solution works	Required for olympiad-style work at Level III; developed from Level I onward.
Has attempted at least one competition or a sample paper	Provides the placement baseline.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Apply named problem-solving strategies deliberately and explain the choice.
- 2 Work with algebraic manipulation, factoring identities, inequalities, and functional equations.
- 3 Apply counting techniques: bijections, complementary counting, casework, recursion, and generating functions.
- 4 Apply the pigeonhole principle, invariants, extremal arguments, and colouring arguments.
- 5 Solve Euclidean geometry problems using angle chasing, similarity, power of a point, and trigonometric methods.
- 6 Apply coordinate, vector, and transformational methods to geometry problems.
- 7 Work with divisibility, modular arithmetic, Diophantine equations, and number-theoretic functions.
- 8 Recognize problem types and select an appropriate approach under time pressure.
- 9 Write complete, rigorous, well-organized solutions.
- 10 Manage competition time, including strategic problem selection and answer verification.
- 11 Persist productively on problems that resist a first approach.

COURSE UNITS

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Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 10-12 LESSONS · TAUGHT FIRST AND REVISITED THROUGHOUT

Problem-Solving Strategy

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TOPICS COVERED

- What makes a competition problem different from a school problem
- Understanding the problem: restating, identifying givens and goals
- Specializing: trying small cases and looking for a pattern
- Generalizing: solving the harder problem because it is easier
- Working backwards from the goal
- Looking for invariants and monovariants
- Extremal principle: consider the largest or smallest
- Symmetry and its exploitation
- Counting two ways as a proof technique
- Proof by contradiction and by contrapositive
- Constructive proof and existence arguments
- Knowing when to abandon an approach
- Writing a solution: structure, completeness, and rigour
- Maintaining a solution journal

LEARNING OBJECTIVES

- 1 Name the strategy being attempted before attempting it.
- 2 Generate small cases systematically and extract a conjecture from them.
- 3 Recognize an invariant in a process problem.
- 4 Write a complete solution that a reader could verify without the student present.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Names a plausible strategy before attempting in 9 of 10 problems.
- Produces systematic small cases in 4 of 5 exploration problems.
- Writes solutions rated complete and rigorous in 4 of 5 journal entries.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The student waits for the method to be recognized rather than exploring.	Exploration is the method. Require small cases to be written before any general attempt — this converts paralysis into activity.

MISCONCEPTION	RECOMMENDED RESPONSE
A solution is considered finished once the answer is found.	The argument is the solution. On olympiad-style problems the answer alone earns little; on short-answer competitions the written argument still trains the reasoning.
Not solving a problem is experienced as failure.	Unsolved problems are the training. State this explicitly and often, and deliberately assign problems beyond the student's current reach.

TEACHING NOTE

This unit is taught first and then never closed — every subsequent lesson opens by naming the strategy under consideration. A student who has internalized eight named strategies has a genuine method for approaching an unfamiliar problem, which is precisely what competition mathematics tests.

UNIT 2 · 14-16 LESSONS**Algebra**

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TOPICS COVERED

- Algebraic manipulation and clever factoring
- Factoring identities: Sophie Germain, difference and sum of powers, symmetric factorizations
- Vieta's formulas and symmetric functions of roots
- Polynomial identities, the Factor and Remainder theorems
- Roots of unity and their applications
- Telescoping sums and products
- Sequences and recursions; solving linear recurrences
- Inequalities: AM-GM, Cauchy-Schwarz, rearrangement, and power mean
- Applying inequalities to optimization problems
- Functional equations and standard substitution techniques
- Systems of nonlinear equations and elimination strategies
- Logarithmic and exponential manipulation
- Complex numbers in algebraic problems

LEARNING OBJECTIVES

- 1 Recognize when a factoring identity applies and use it to simplify a hard expression.
- 2 Apply Vieta's formulas to relate roots and coefficients without solving.
- 3 Select and apply an appropriate classical inequality with the equality condition stated.
- 4 Solve a functional equation using a systematic substitution strategy.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Recognizes applicable identities in 8 of 10 problems.
- Applies Vieta's formulas correctly in 9 of 10 trials.
- Selects an appropriate inequality with correct equality conditions in 4 of 5 problems.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Roots are computed when only symmetric functions of them are needed.	Vieta's formulas give the symmetric functions directly. Computing the roots is usually the slow path and sometimes impossible.
AM-GM is applied without checking the equality condition.	The equality condition is often the point of the problem. State it every time.
Functional equations are attacked by guessing the function.	Substitute strategically — zero, one, x for y , symmetry — and derive properties before conjecturing.

TEACHING NOTE

Vieta's formulas and the factoring identities are the highest-yield algebraic tools at AMC and AIME level. A student fluent in these solves a meaningful fraction of the algebra problems on any given paper substantially faster than one who is not.

UNIT 3 · 14-16 LESSONS

Combinatorics

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TOPICS COVERED

- The addition and multiplication principles
- Permutations, combinations, and multisets
- Complementary counting and casework
- Bijections and counting one set by counting another
- Stars and bars and distributions
- The principle of inclusion-exclusion
- Recursions in counting problems
- The binomial theorem and combinatorial identities
- Counting two ways as a proof technique
- The pigeonhole principle, simple and generalized
- Invariants and colouring arguments
- Extremal combinatorics
- Introductory graph theory: paths, cycles, connectivity, trees
- Probability as counting; expected value and linearity of expectation
- Generating functions at an introductory level
- Games and winning strategies

LEARNING OBJECTIVES

- 1 Set up a count by an explicit bijection and justify that it is a bijection.
- 2 Choose between direct counting, complementary counting, and inclusion-exclusion.
- 3 Apply the pigeonhole principle by identifying the pigeons and the holes explicitly.
- 4 Use linearity of expectation to avoid a difficult direct computation.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Selects an efficient counting approach in 8 of 10 problems.
- Applies inclusion-exclusion correctly in 9 of 10 trials.
- Identifies pigeons and holes correctly in 4 of 5 pigeonhole problems.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Overcounting occurs because indistinguishable arrangements are counted separately.	Ask explicitly what is being counted and whether two arrangements are the same. Overcounting is the dominant error in combinatorics.
The pigeonhole principle is invoked without identifying the objects and the categories.	State both explicitly. If they cannot be stated, the principle does not apply yet.
Expected value is computed by finding the full distribution.	Linearity of expectation frequently avoids this entirely and does not require independence.

TEACHING NOTE

Combinatorics is where the strongest competitors are made, because it rewards creative reframing more than accumulated technique. It is also where the strategy vocabulary of Unit 1 is used most heavily — connect the two explicitly and often.

UNIT 4 · 14-16 LESSONS

Geometry

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TOPICS COVERED

- Angle chasing: the fundamental technique
- Similar triangles and proportional reasoning
- The power of a point theorem
- Cyclic quadrilaterals and the conditions for concyclicity
- Ptolemy's theorem
- The incircle, circumcircle, and the standard triangle centres
- Ceva's and Menelaus's theorems
- Area methods and the ratio of areas
- The Law of Sines and Cosines in competition contexts
- Trigonometric identities in geometric problems
- Coordinate geometry as a computational method
- Vectors and complex numbers as geometric tools
- Transformations: reflection, rotation, homothety, inversion introduced
- Three-dimensional geometry problems
- Constructing an accurate diagram and what it reveals

LEARNING OBJECTIVES

- 1 Solve a problem by systematic angle chasing with all angles labelled.

- 2 Recognize a concyclicity condition and exploit it.
- 3 Choose deliberately among synthetic, coordinate, trigonometric, and complex number methods.
- 4 Draw an accurate diagram and use it to generate a conjecture.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves angle chasing problems in 4 of 5 attempts.
- Recognizes cyclic quadrilaterals in 8 of 10 applicable problems.
- Selects an efficient method in 8 of 10 problems when compared with the teacher's choice.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
An inaccurate diagram is drawn, and a false conjecture follows from it.	Use a ruler and compass, or dynamic geometry software. An accurate diagram is a source of true conjectures; a sloppy one is a source of wasted hours.
Coordinates are used where a synthetic argument would be far shorter.	Coordinates always work and are frequently slow. Estimate the cost of each method before committing.
Configuration dependence is ignored — a solution is given for one diagram only.	Check whether the argument survives when the point moves. This is a genuine rigour issue at olympiad level.

TEACHING NOTE

Angle chasing is the foundational technique and the one that most rewards patient practice. Require every angle in the diagram to be labelled before any theorem is invoked — most competition geometry problems yield to complete labelling alone.

UNIT 5 · 12-14 LESSONS

Number Theory

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TOPICS COVERED

- Divisibility, the division algorithm, and the Euclidean algorithm
- Prime factorization and the fundamental theorem of arithmetic
- GCD and LCM and their properties
- Modular arithmetic and congruences
- Solving linear congruences and the Chinese Remainder Theorem
- Fermat's little theorem and Euler's theorem
- Wilson's theorem
- Orders and primitive roots introduced
- The number and sum of divisors functions
- Diophantine equations: linear and Pythagorean
- Infinite descent
- Base representations and digit problems
- Floor and ceiling functions
- Quadratic residues introduced
- Number-theoretic bounding arguments

LEARNING OBJECTIVES

- 1 Apply modular arithmetic to determine remainders and last digits efficiently.
- 2 Use the Chinese Remainder Theorem to combine congruences.
- 3 Solve a linear Diophantine equation completely, giving all solutions.
- 4 Apply a bounding argument to limit the cases requiring examination.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Applies modular arithmetic correctly in 9 of 10 problems.
- Solves linear Diophantine equations completely in 9 of 10 trials.
- Uses bounding to reduce case counts in 4 of 5 applicable problems.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Division is performed in modular arithmetic without checking invertibility.	Division requires the divisor to be coprime to the modulus. Multiply by the inverse instead, and verify it exists.
A Diophantine equation is answered with one solution rather than all.	Give the complete family. Partial answers are incomplete answers.
Every case is examined when bounding would eliminate most of them.	Bound first, then enumerate. Bounding is the characteristic number theory technique.

TEACHING NOTE

Number theory is the most self-contained of the four areas and the one where a motivated student can advance fastest. It is also disproportionately represented at AIME level relative to its presence in school curricula.

UNIT 6 • 10-12 LESSONS PLUS SCHEDULED PRACTICE CONTESTS**Competition Practice and Contest Strategy**

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TOPICS COVERED

- The competition landscape: AMC 8, AMC 10, AMC 12, AIME, and beyond
- Scoring structures and what they imply for guessing
- Time allocation across a paper
- Strategic problem selection: which to attempt and in what order
- Recognizing a problem type quickly
- Answer verification techniques: estimation, special cases, dimensional checks
- Managing the psychology of a difficult problem mid-contest
- Full-length timed practice contests
- Post-contest analysis: what was missed and why
- Building and maintaining a personal problem journal
- Writing up solutions for olympiad-style grading
- Preparing for team competitions and collaborative problem-solving
- Longer-term training planning across multiple years

LEARNING OBJECTIVES

- 1 Allocate time strategically across a full contest paper.
- 2 Decide within one minute whether to attempt, defer, or skip a problem.
- 3 Verify an answer by an independent method before committing.
- 4 Analyze a completed contest and derive a specific training plan from it.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Completes practice contests with appropriate time allocation in 3 consecutive attempts.
- Verifies answers independently in 8 of 10 attempted problems.
- Produces a specific, actionable training plan from a contest analysis.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Problems are attempted in strict numerical order.	Difficulty generally increases but not monotonically. Scan the paper first and attempt the accessible problems wherever they appear.
Too long is spent on one problem, at the cost of several easier ones.	Set a personal time limit per problem and enforce it. Return if time permits.

MISCONCEPTION	RECOMMENDED RESPONSE
Post-contest analysis is limited to reading the official solutions.	Reattempt the missed problems before reading anything. The reattempt is where the learning happens; reading a solution feels productive and mostly is not.

TEACHING NOTE

Reattempting a missed problem before reading the solution is the highest-value practice in competition training, and it is the one students most resist. Make it a required step in the problem journal, not a suggestion.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	A competition-style paper appropriate to the student's school level; determines level placement and identifies which of the four areas is weakest.
Problem Journal	Continuous	Every problem attempted, with the strategy named, the work shown including failed approaches, and a written solution. The central document of the course and the primary evidence of growth.
Unit Problem Set	End of each unit	A take-home set of eight to twelve problems of graduated difficulty, including at least two the student is not expected to solve.
Timed Practice Contest	Every 3–4 weeks	A full-length past paper under authentic conditions, followed by reattempt-before-solutions analysis.
Write-Up Review	Monthly	A single solution written to olympiad standard and critiqued for completeness, rigour, and organization.
Level Advancement Assessment	End of course	Determines readiness for the next level; combines a timed paper with a written proof component.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Level I — AMC 8 track	1 lesson/week, 75 min	Typically Grades 5–8; approximately 60 lessons over one to two years.
Level II — AMC 10/12 track	1–2 lessons/week, 90 min	Typically Grades 8–11; approximately 75 lessons.
Level III — AIME and olympiad track	2 lessons/week, 90 min	Typically Grades 9–12 with a qualifying score; approximately 90 lessons with heavy independent work.
Contest Intensive	2 lessons/week, 6 weeks	Before a specific competition date; timed papers and targeted area work.
Summer Program	3 lessons/week, 6 weeks	One area in depth — most commonly combinatorics or number theory, which schools cover least.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Sixty to ninety minutes per lesson, and the structure differs fundamentally from every other course in the catalog. Assignments contain few problems — typically four to eight — and each is expected to take real time. Students record every attempt in the problem journal, including approaches that failed and why they were abandoned; the failed approaches are as informative to the teacher as the successful ones. At least one problem per assignment is chosen to be beyond the student's current reach, and they are told so, because learning to work productively on an unsolved problem is the actual curriculum. Solutions are written in full prose, not as chains of equations.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Names a strategy before attempting a problem.
- ◆ Explores systematically with small cases and extracts conjectures.
- ◆ Writes complete solutions a reader can verify independently.
- ◆ Applies factoring identities and Vieta's formulas fluently.
- ◆ Selects and applies classical inequalities with equality conditions.
- ◆ Counts by bijection, complementary counting, and inclusion-exclusion.
- ◆ Applies the pigeonhole principle, invariants, and extremal arguments.
- ◆ Uses linearity of expectation to avoid direct computation.
- ◆ Solves geometry problems by systematic angle chasing.

- ◆ Recognizes cyclic quadrilaterals and applies power of a point.
- ◆ Chooses deliberately among synthetic, coordinate, and complex methods.
- ◆ Applies modular arithmetic, CRT, and bounding arguments.
- ◆ Solves Diophantine equations completely.
- ◆ Allocates time strategically across a full contest paper.
- ◆ Persists productively on problems that resist a first approach.
- ◆ Qualified for the next competition level.

MATERIALS AND RESOURCES

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- The Art of Problem Solving series, matched to the student's level.
- Past papers: AMC 8, AMC 10, AMC 12, and AIME, with a substantial archive.
- Dynamic geometry software for accurate diagrams and conjecture generation.
- Վարժարան problem journal template — the central document of the course.
- Վարժարան competition problem bank — approximately 3,000 problems tagged by area, technique, difficulty, and the strategy that unlocks them.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.